

# Bangladesh University of Business and Technology (BUBT)



## **LAB Report**

Course Code :  
Course Title :  
Experiment No. :  
Experiment Name :

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Intake : 53  
Section : 01  
Program : B.Sc. in CSE

Name:

Dept. of  
Bangladesh University of  
Business and Technology

**Date of Submission :**

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**Signature of Teacher**

## **Experiment Name:** Network Cabling

### **Introduction:**

Network cabling connects computers and other network devices so they can communicate. It is essential in any network. In this experiment, we learned to make straight-through and crossover Ethernet cables, understanding how physical connections are implemented in real networks.

### **Tools and Equipment:**

1. UTP (Ethernet) cable
2. RJ-45 connectors
3. Crimping tool
4. Cable cutter and stripper
5. LAN cable tester

### **Procedure:**

#### **Section 1: Making a Straight-through Ethernet Cable**

1. Take a UTP cable and cut it to the required length.
2. Strip about 1–2 inches of the outer insulation from one end.
3. Separate the eight wires and straighten them.
4. Arrange the wires according to the T568B standard.
5. Trim the wires evenly to ensure proper insertion.
6. Insert the wires carefully into the RJ-45 connector.
7. Ensure all wires reach the end of the connector.
8. Crimp the connector firmly using the crimping tool.
9. Repeat steps 2–8 for the other end using the same T568B wiring order.
10. Test the cable using a LAN tester to ensure all connections are correct.

#### **Section 2: Making a Crossover Ethernet Cable**

1. Take another UTP cable and cut it to the required length.
2. Strip about 1–2 inches of the outer insulation from one end.
3. Separate the eight wires and straighten them.
4. Arrange the wires according to the T568A standard on one end.
5. Trim the wires evenly and insert them into an RJ-45 connector.
6. Crimp the connector firmly using the crimping tool.
7. Strip the other end of the cable and straighten the wires.
8. Arrange the wires according to the T568B standard on the other end.
9. Insert the wires into the RJ-45 connector, crimp firmly, and check that all wires are secure.

10. Test the cable with a LAN tester. Verify that the tester shows a crossed light pattern, confirming correct crossover wiring.

## **Conclusion:**

From this experiment, we learned how to make both straight-through and crossover Ethernet cables. We understood the differences between T568A and T568B standards and the importance of correct wire sequence. This experiment provided practical hands-on experience in creating and testing network cables used in real-world networks.